Associative Memory & Fast Weights: Concept Summary

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When stored keys are not orthogonal, a linear fast-weight state adds non-target contributions to retrieval; in our synthetic setup, increasing controlled key overlap increases measured cross-talk. This central falsifiable claim establishes both the algebraic mechanism of memory degradation in un-gated outer-product recurrence and an empirical observation of how key geometric alignment degrades recall in finite-dimensional state representations.

1. Design Motivation: The KV-Cache Bottleneck

Conventional autoregressive Transformers compute self-attention using $\mathrm{Softmax}(QK^T)V$, requiring continuous storage of all past key and value vectors. This Key-Value (KV) cache grows linearly with sequence length ($O(N \cdot d)$), increasing memory-bandwidth pressure during inference. Fast-weight architectures (Schlag et al., ICML 2021) and linear recurrent models (Beck et al., NeurIPS 2024; Gu & Dao, 2023) address this bottleneck by compressing unbounded context into a fixed-size recurrent state matrix $S_t \in \mathbb{R}^{d \times d}$.

2. Technical Mechanism and Observed Trade-Off

Instead of storing tokens sequentially, fast-weight memory accumulates key-value pairs (k_t, v_t) via outer products decayed by retention factor $\lambda \in (0, 1]$:

$$S_t = \lambda S_{t-1} + v_t k_t^T = \sum_{i=1}^{t} \lambda^{t-i} v_i k_i^T$$

Associative retrieval is executed via a single matrix-vector product $y_t = S_t q$. For a query targeting stored association j ($q = k_j$), the output decomposes algebraically into:

$$y_t = \underbrace{\lambda^{t-j} v_j (k_j^T q)}_{\text{Target Contribution}} + \underbrace{\sum_{i \neq j} \lambda^{t-i} v_i (k_i^T q)}_{\text{Cross-Talk Contribution}}$$

The core trade-off is constant $O(1)$ memory per step vs. retrieval interference: because the state matrix has algebraic rank bounded by d ($\mathrm{Rank}(S) \leq d$), stored items cannot all occupy orthogonal subspaces once sequence length $N > d$. Overlapping keys produce non-zero projections ($k_i^T q \neq 0$), injecting non-target values additively into the readout.

3. Comparison Across Architectural Paradigms

Dimension	Standard Softmax Attention	Linear Fast Weights / DeltaNet	BDH Synaptic Architecture
Memory Footprint	$O(N \cdot d)$ (Grows linearly)	$O(d^2)$ (Fixed recurrent matrix)	$O(d^2)$ (Sparse synaptic matrix)
Inference Step Compute	$O(N \cdot d)$ compute per token	$O(d^2)$ matrix-vector product	$O(k \cdot d)$ sparse activations ($k \ll d$)
Cross-Talk Behavior	Suppressed exponentially by softmax	Additive linear superposition	Sparse positive teaching abstraction changes which interactions contribute
Interpretability	Dense $N \times N$ token attention matrix	Superimposed continuous weights	Sparse neural circuits
Scaling / State Capacity	Hardware HBM bandwidth	State rank $\mathrm{Rank}(S) \leq d$	Constrained by state dimension and active support

4. Empirical Evidence: Strengths and Weaknesses

- **Demonstrated Advantage:** The recurrent formulation keeps an explicit state whose dimensions do not grow with the number of processed tokens, matching full causal attention on associative recall when key representations are mutually orthogonal (Schlag et al., 2021 `[BENCHMARK EVALUATION]`).
- **Weaknesses and Untested Regimes:** On high-entropy tasks requiring precise recall of hundreds of non-orthogonal entities, [OUR EXPERIMENT] Synthetic sweeps at $d = 8$ show measured retrieval degradation as controlled key overlap increases (see also Sun et al., 2024 for architectural discussions of recurrent hidden-state capacity). Sweeps at $d = 8$ (50 Monte Carlo trials per configuration) show mean cosine similarity dropping from 1.0000 ($\rho = 0.0$) to 0.7647 ($\rho = 0.45$), with higher synthetic memory load ($N/d = 1.5$) further elevating measured error to mean cosine similarity 0.5764 `[SYNTHETIC BENCHMARK]`.

5. Role of BDH and BDH-CQ

- **Dragon Hatchling (BDH):** Proposed by Kosowski et al. (2025; arXiv:2509.26507), BDH provides a contrasting research architecture centered on sparse positive activations ($a \geq 0$) and local synaptic plasticity. This project uses a simplified single-layer abstraction to explore that conceptual contrast `[TEACHING ABSTRACTION]`.
- **BDH-CQ Context:** Engdahl et al. (2026; arXiv:2608.09888) explores in-context demonstration learning where input-output demonstrations update recurrent synaptic states for multi-step latent reasoning without emitting intermediate tokens `[RESEARCH CONTEXT]`.

6. Primary Open Limitation

The central open question in fast-weight memory is representational expressivity under finite-dimensional bounds: while non-negative sparsity alters which active interactions contribute, a fixed $d \times d$ synaptic state remains constrained by the geometry of $\mathbb{R}^d$. Understanding the exact trade-off between synaptic sparsity and representational expressivity under natural language token distributions remains an open research frontier.